Semantic Field Overlap

KWTA applied to word activation maps.

Store a dataset of 1.8M word associations (e.g. "ship"→"sail") in hetero-associative memory. Retrieve activation maps for two input tokens and compute their overlap using kWTA.

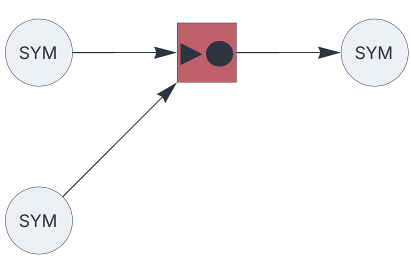
```
<< "Circuits.m"
```

Hetero-associative memory circuit

```
hyperparameters = {5000,10};
```

```
dataflow = {
        input[codec] [1] [],
        input[codec] [2] [],
        associator[] [11] [2, 1],
        output[codec] [] [11],
        _ → hyperparameters
    };
```

```
c = Circuit[dataflow]; c[Graph, ImageSize→ 220]
```

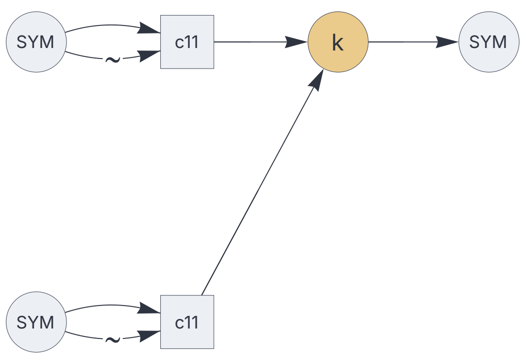


Testing circuit

Embedding the memory circuit above.

```
testcircuit = {
         input[codec] [11] [],
        input[codec] [12] [],
        circuit[c][21][11, 11"~"], (* "~" clears data *)
        circuit[c][22][12, 12"~"],
        kwta[][31][{21,22}],
        output[codec] [] [31],
        _ → hyperparameters,
        "multiplex"→"10" (* One empty input to flush the delay. *)
    };
```

```
test = Circuit[testcircuit]; test[Graph, ImageSize→ 280]
```



Import dataset

```
data = Import[ NotebookDirectory[] <>  "/data/word-associations.csv.zip",
          "word-associations.csv"];
```

```
{hobbyists ⟷ collectors, ballpark ⟷ diamond, sighting ⟷ appearance,
 superfine ⟷ wool, sensorimotor ⟷ control, sneaking ⟷ tiptoe,
 microclimate ⟷ ecosystem, patternless ⟷ unordered,
 totality ⟷ perfectness, squalidly ⟷ sordid}
```

1 888 505

Train

```
c @@@ data;
```

Test

Retrieve and decode semantic fields:

```
c[ "ship", Null]
```

```
Sequence[{harbor, mast, stern, bow, deck, helm, nautical,
  ocean, marine, port, starboard, navigation, maritime, boat,
  vessel, crew, sail, voyage, launch, sea, anchor, fleet, seas,
  captain, sailing, compass, cargo, dock, hull, shipyard, keel}]
```

```
c["boat", Null]
```

```
Sequence[{trip, harbor, mast, stern, bow, deck, nautical, fishing, ocean,
  maritime, river, craft, water, vessel, ferry, crew, passenger, sail,
  ride, voyage, sea, anchor, owner, channel, pier, motor, speed, navigate,
  ramp, seas, wave, engine, captain, sailing, raft, tour, rudder, buoyancy,
  lake, yacht, dock, knot, row, canoe, hull, mooring, builder, paddle,
  stumped, marina, buoy, wharf, keel, oar, dinghy, regatta, lifeboat}]
```

Compute kWTA, using the testing circuit.

```
test["ship", "boat"]
```

```
Sequence[{harbor, mast, stern, bow, deck, nautical, ocean, maritime, vessel,
  crew, sail, voyage, sea, anchor, seas, captain, sailing, dock, hull, keel}]
```